

Instructions:

You are a SOC Analyst at Global Logistics. Emily Nguyen, a member of the marketing team, recently had trouble signing into her Dropbox account after trying to access it for the first time in months and reached out to her manager for assistance. The next day, she received an email that claims a password change request was made for her Dropbox account. The email includes a link for resetting her password, but Emily is unsure if the request is legitimate. Concerned about potential phishing, she has forwarded the email to the security team for analysis.

Using what you've learned within this domain, perform a detailed email analysis on the challenge2.eml file to answer the report questions below.

Q1 Based on the contents of the email header, what is the full date and time of the email delivery?

Sun, 12 May 2024 04:10:52 +0000

Q2 What is the subject of the email?

Reset your Dropbox password

Q3 Who was the email sent to?

emily.nguyen@glbllogistics.co

Q4 Based on the sender's display name, who does the email claim to be from?

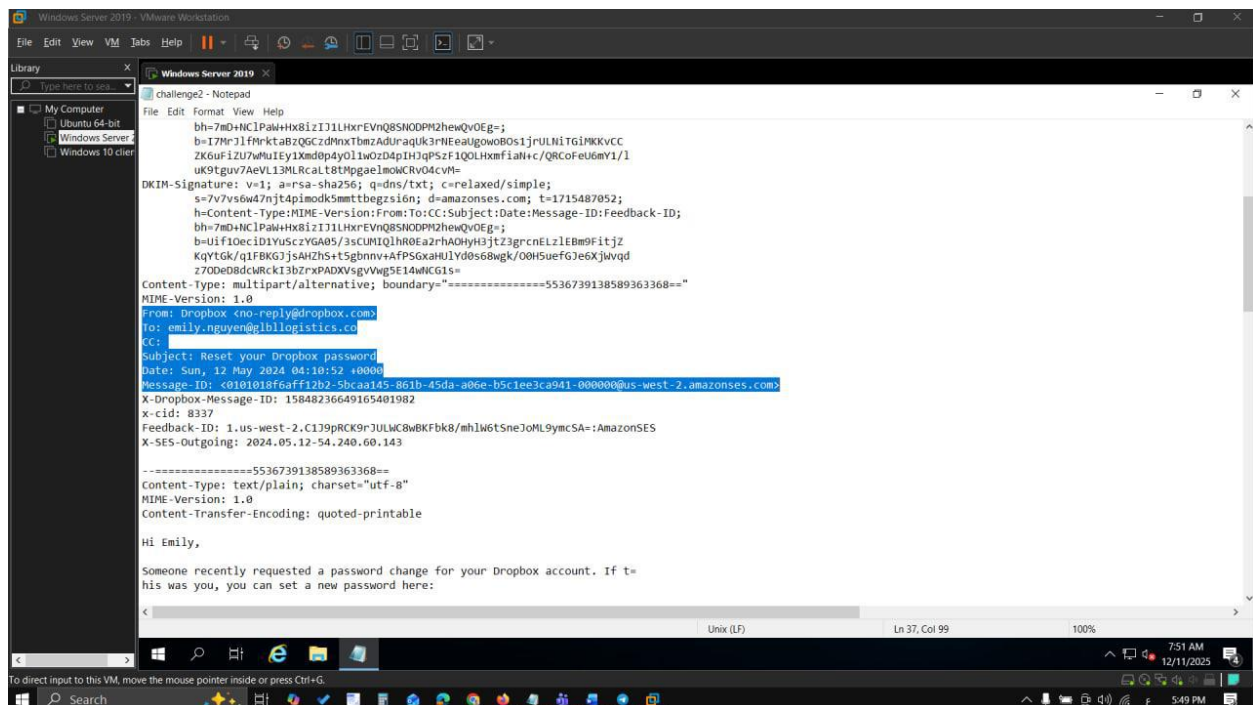
Dropbox

Q5 What is the sender's email address?

no-reply@dropbox.com

Q6 What email address is used for receiving bounced emails?

0101018f6aff12b2-5bcaa145-861b-45da-a06e-b5c1ee3ca941-000000@email.dropbox.com

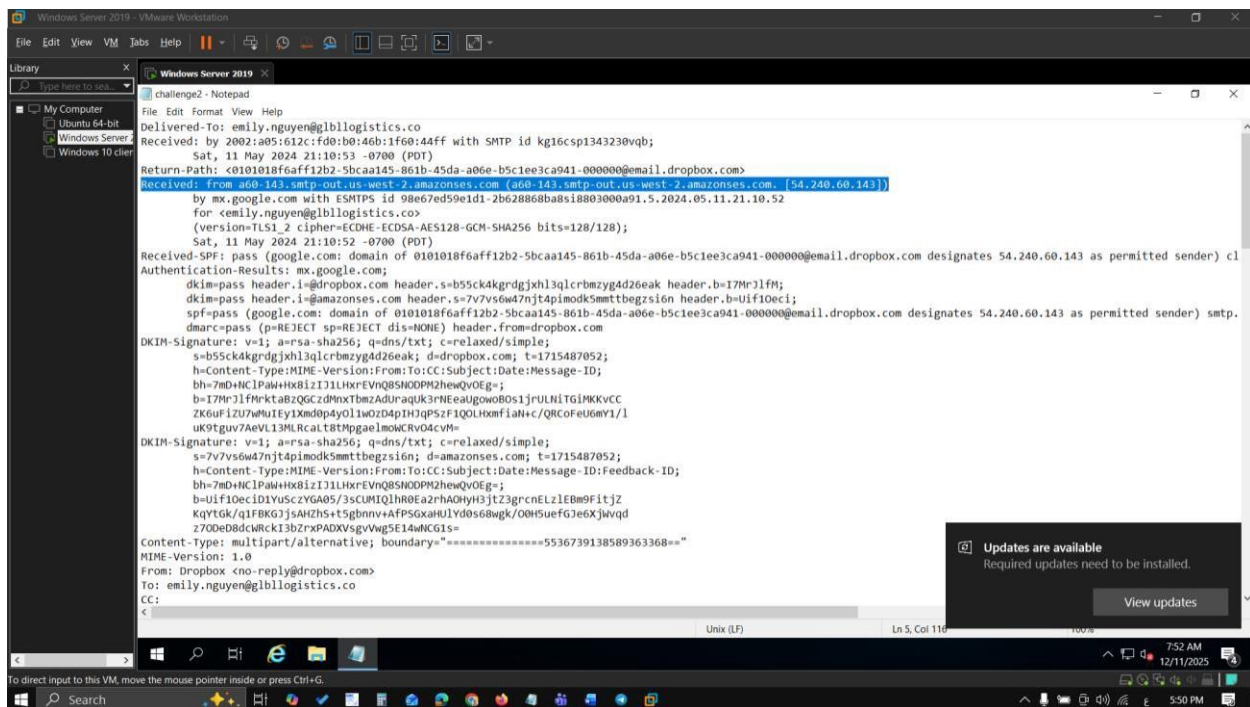


The screenshot shows a Windows Server 2019 virtual machine environment. A Notepad window titled 'challenge2 - Notepad' is open, displaying the raw MIME content of an email. The email header includes the following fields: 'From: Dropbox <no-reply@dropbox.com>', 'To: emily.nguyen@glbllogistics.co', 'Subject: Reset your Dropbox password', 'Date: Sun, 12 May 2024 04:10:52 +0000', 'Message-ID: <0101018f6aff12b2-5bcaa145-861b-45da-a06e-b5c1ee3ca941-000000@us-west-2.amazonaws.com>', 'X-Dropbox-Message-ID: 15848236649165401982', 'x-cid: 8337', 'Feedback-ID: 1.us-west-2.C139pRCK9rJULWC8wBKFbK8/mh1w6tSneJoML9ymcSA=:AmazonSES', and 'X-SES-Outgoing: 2024.05.12-54.240.60.143'. The body of the email begins with 'Hi Emily,' followed by a message about a password change request for a Dropbox account.

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File Edit Format View Help
challenge2 - Notepad
bh=7mD+NC1Pam+Hx81zI1LHxrEVnQ8SNODPM2hewQvOEG=;
b=I7M+1JfMrktaBzQGczdMnxTbmZAdUraqUk3rNEeaUgowoBos1jrULNiTG1MKKvCC
ZK6uF1ZU7wMuEy1Xmd0p4y01woZD4p1H3qP5zF1QOLHxmfiAn+c/QRCofeU6mY1/1
uk9Tguv7AeVL13MLRcaLt8TpggaeImowCRvO4cvH=
DKIM-Signature: v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple;
s=7v7vs6w47njt4pimodk5mmttbgzsi6n; d=amazonse.com; t=1715487052;
h=Content-Type:MIME-Version:From:To:CC:Subject:Date:Message-ID:Feedback-ID;
bh=7mD+NC1Pam+Hx81zI1LHxrEVnQ8SNODPM2hewQvOEG=;
b=UifioecID1YuSczVGA05/3scUMIQ1hr0Ea2rhaOHYH3jtZ3grcnELzLEBm9FitjZ
KqYtGk/q1FBKG7jsAHZHS+t5gbnnv+AFPSGxahULY0s68wgk/O0HSuefGje6Xjwvqd
z70DeDBdcWRckI3bZrxPADXVsgvVwg5E14wNCG1s=
Content-Type: multipart/alternative; boundary="=====5536739138589363368=="
MIME-Version: 1.0
From: Dropbox <no-reply@dropbox.com>
To: emily.nguyen@glbllogistics.co
CC:
Subject: Reset your Dropbox password
Date: Sun, 12 May 2024 04:10:52 +0000
Message-ID: <0101018f6aff12b2-5bcaa145-861b-45da-a06e-b5c1ee3ca941-000000@us-west-2.amazonaws.com>
X-Dropbox-Message-ID: 15848236649165401982
x-cid: 8337
Feedback-ID: 1.us-west-2.C139pRCK9rJULWC8wBKFbK8/mh1w6tSneJoML9ymcSA=:AmazonSES
X-SES-Outgoing: 2024.05.12-54.240.60.143
=====5536739138589363368==
Content-Type: text/plain; charset="utf-8"
MIME-Version: 1.0
Content-Transfer-Encoding: quoted-printable

Hi Emily,

Someone recently requested a password change for your Dropbox account. If t=
his was you, you can set a new password here:
```



Q9 What is the Autonomous System Number (ASN) that owns this IP address?

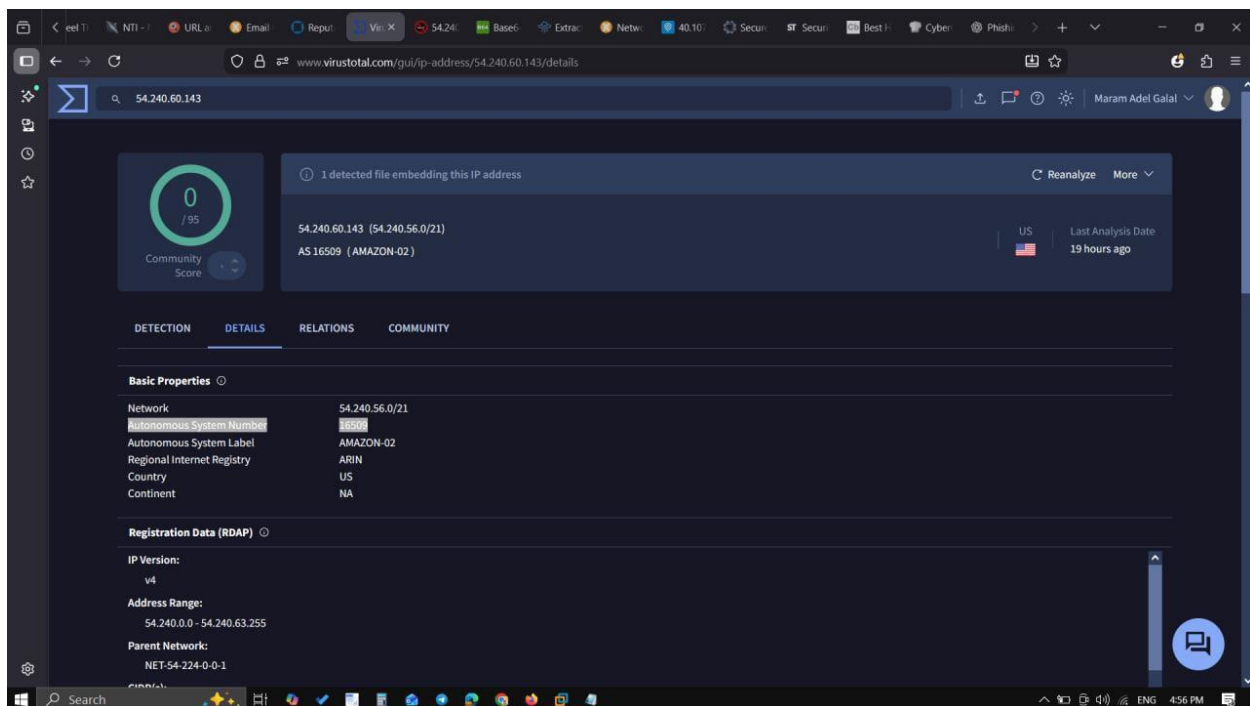
AS16509

Q7 What is the IP address of the sender's email server?

54.240.60.143

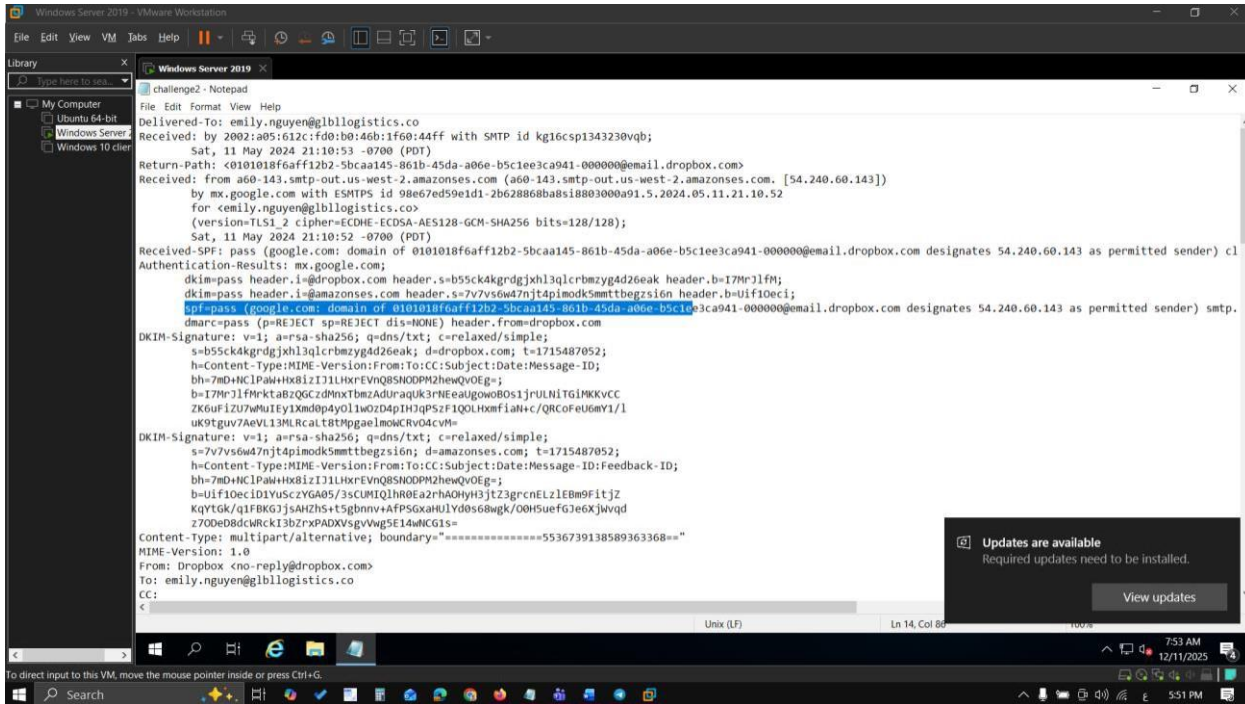
Q8 What is the resolved hostname of the sender's IP address?

a60-143.smtp-out.us-west-2.amazonaws.com



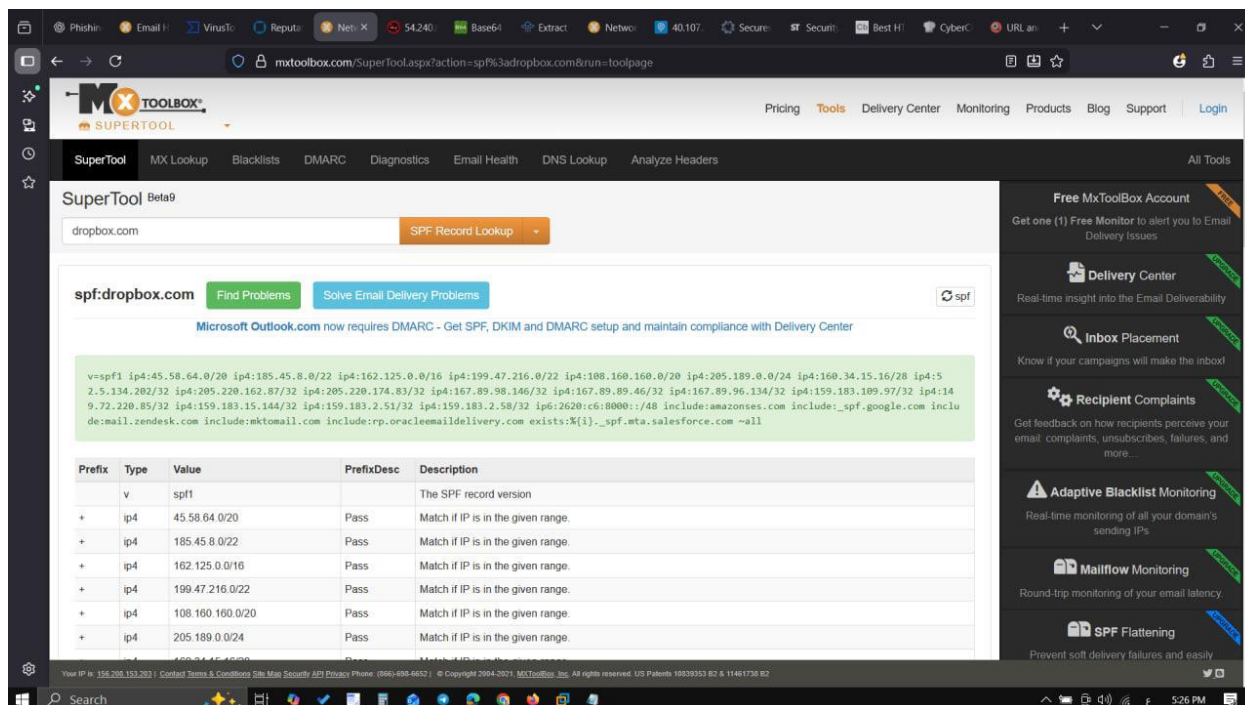
Q10 What was the result of the SPF check?

Pass



Q11 What is the full SPF record of the sender's domain?

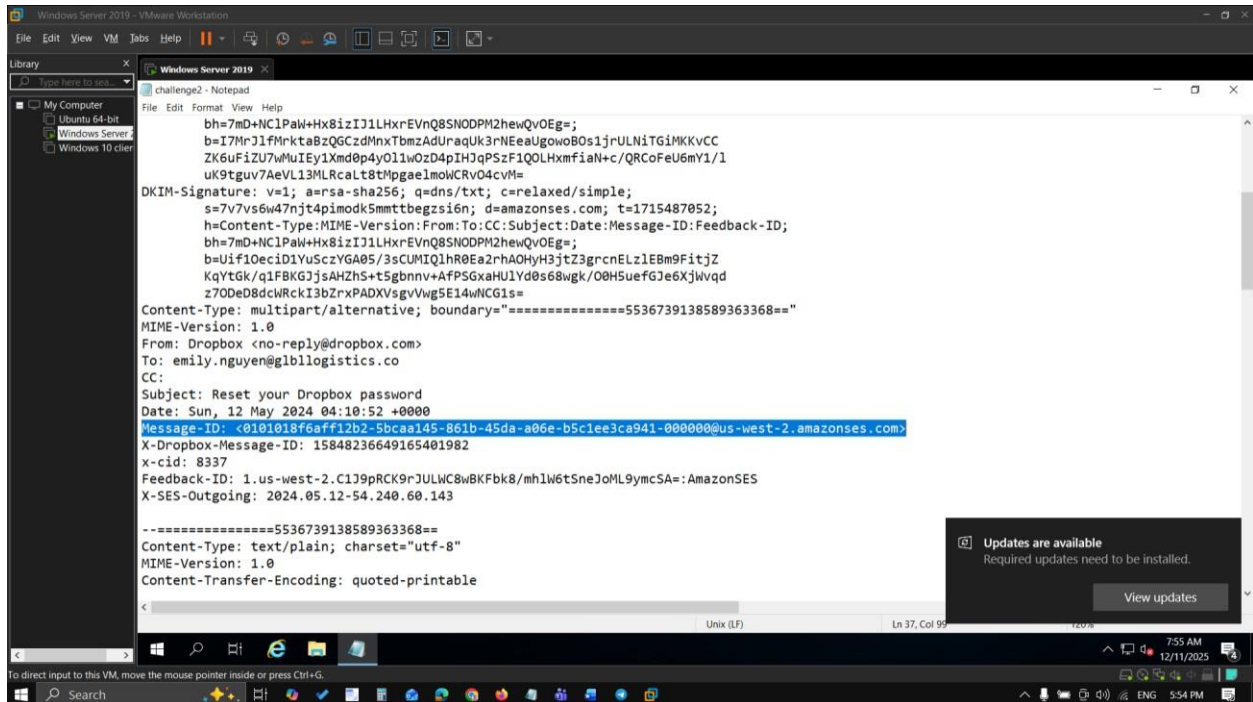
v=spf1 include:amazonses.com -all



Q12 What is email's Message ID?

0101018f6aff12b2-5bcaa145-861b-45da-a06e-b5c1ee3ca941-000000@us-west-

2.amazonses.com



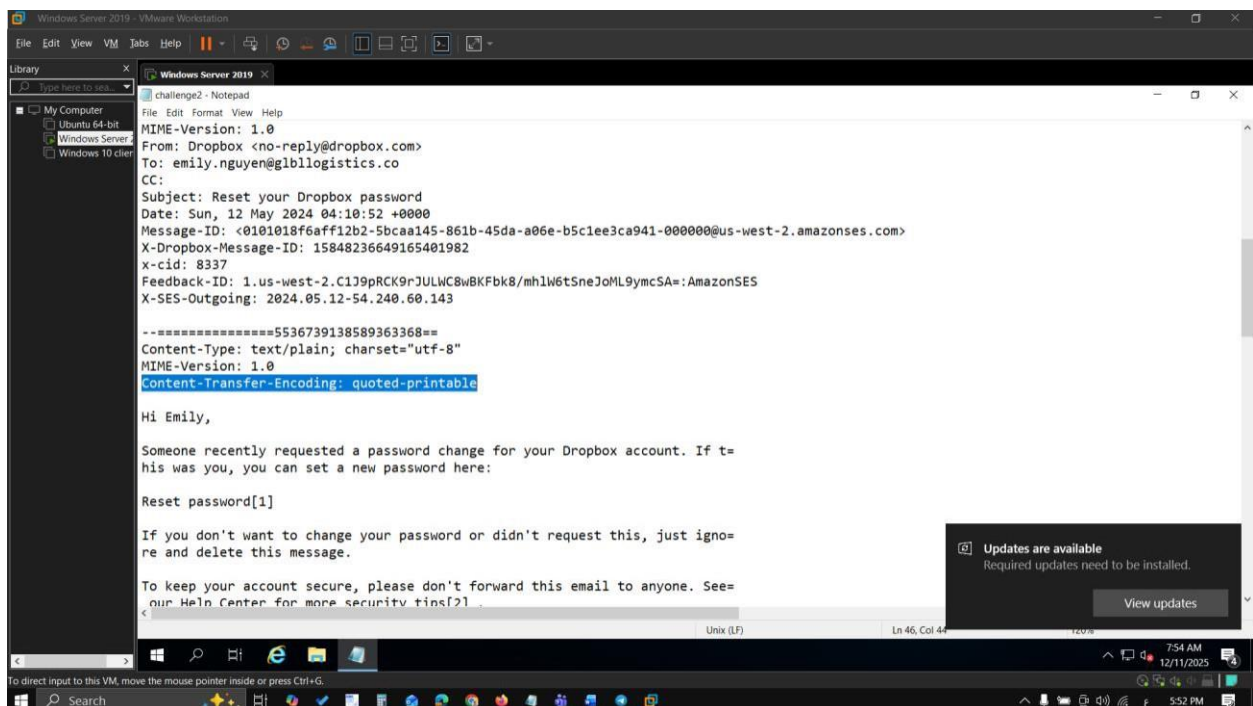
```
challenge2 - Notepad
File Edit Format View Help

bh=7mD+NC1PaW+Hx8izIj1LHxrEVnQ8SNODPM2hewQvOEg=;
b=I7MrJ1fMktaBzQGcdMnxTbmzAdUraqUk3rNEeaUgowoB0s1jrULNiTGIMKvCC
ZK6uFiZU7WuIEy1Xmd0p4yO1lwOzD4pIHJqPSzF1QOLHxmfiAN+c/QRCoFeU6mY1/1
uk9tguv7AeVL13MLRcalt8tMpgaelmowCRvO4cvM=
DKIM-Signature: v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple;
s=7v7vs6w47njt4pimodk5mmttbegzsi6n; d=amazonses.com; t=1715487052;
h=Content-Type:MIME-Version:From:To:CC:Subject:Date:Message-ID:Feedback-ID;
bh=7mD+NC1PaW+Hx8izIj1LHxrEVnQ8SNODPM2hewQvOEg=;
b=UifiOecid1YuSczYGA05/3sCUMIQ1hr0Ea2rhAOHyH3jtZ3grcnELzIEBm9FitjZ
KqYtGk/q1FBKGJjsAHZhs+t5gbnnv+AfPSGxaHULYd0s68wgk/00H5uefGJe6XjWvqd
z70DeD8dclRckI3bZrxPADXVsgvVvg5E14wNCG1s=
Content-Type: multipart/alternative; boundary="=====5536739138589363368=="
MIME-Version: 1.0
From: Dropbox <no-reply@dropbox.com>
To: emily.nguyen@glbllogistics.co
CC:
Subject: Reset your Dropbox password
Date: Sun, 12 May 2024 04:10:52 +0000
Message-ID: <0101018f6aff12b2-5bcaa145-861b-45da-a06e-b5c1ee3ca941-000000@us-west-2.amazonses.com>
X-Dropbox-Message-ID: 15848236649165401982
x-cid: 8337
Feedback-ID: 1.us-west-2.C1j9pRCK9rJULWC8wBKFbk8/mhlw6tSneJoML9ymcSA=:AmazonSES
X-SES-Outgoing: 2024.05.12-54.240.60.143

-----5536739138589363368==
Content-Type: text/plain; charset="utf-8"
MIME-Version: 1.0
Content-Transfer-Encoding: quoted-printable
```

Q13 What type of encoding was used to transfer the email body content?

quoted-printable



```
challenge2 - Notepad
File Edit Format View Help

MIME-Version: 1.0
From: Dropbox <no-reply@dropbox.com>
To: emily.nguyen@glbllogistics.co
CC:
Subject: Reset your Dropbox password
Date: Sun, 12 May 2024 04:10:52 +0000
Message-ID: <0101018f6aff12b2-5bcaa145-861b-45da-a06e-b5c1ee3ca941-000000@us-west-2.amazonses.com>
X-Dropbox-Message-ID: 15848236649165401982
x-cid: 8337
Feedback-ID: 1.us-west-2.C1j9pRCK9rJULWC8wBKFbk8/mhlw6tSneJoML9ymcSA=:AmazonSES
X-SES-Outgoing: 2024.05.12-54.240.60.143

-----5536739138589363368==
Content-Type: text/plain; charset="utf-8"
MIME-Version: 1.0
Content-Transfer-Encoding: quoted-printable

Hi Emily,

Someone recently requested a password change for your Dropbox account. If t=
his was you, you can set a new password here:

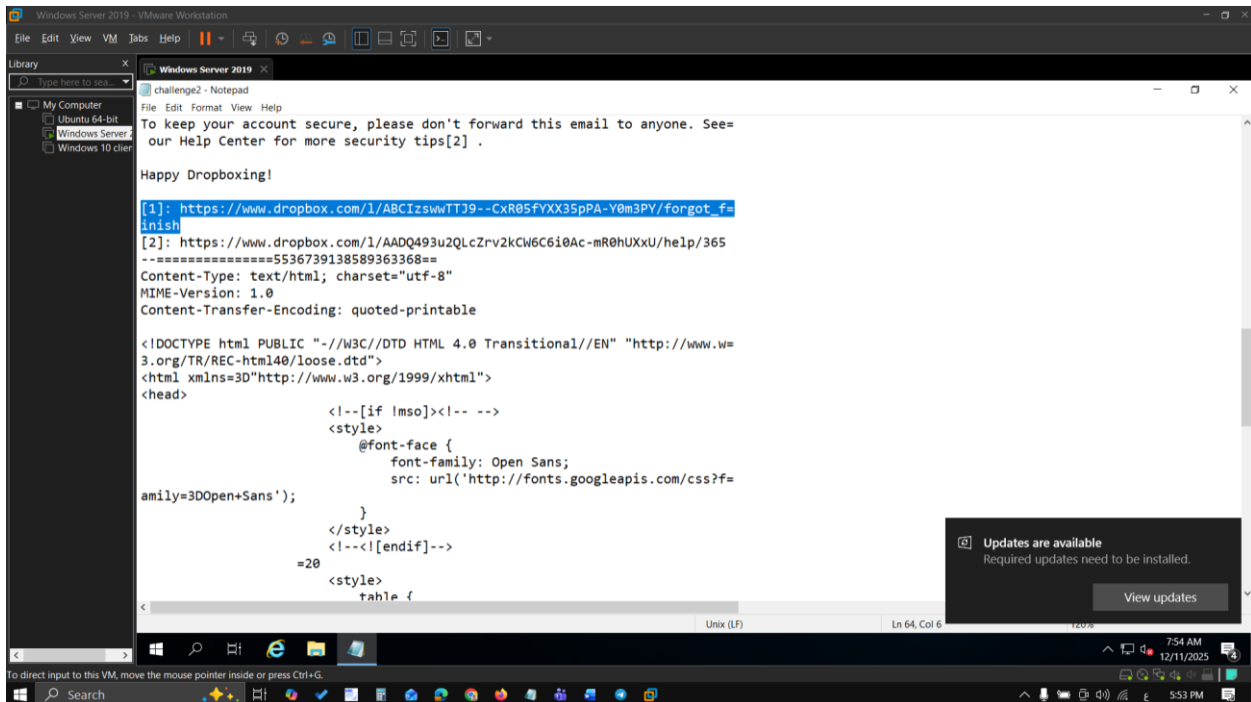
Reset password[1]

If you don't want to change your password or didn't request this, just igno=
re and delete this message.

To keep your account secure, please don't forward this email to anyone. See=
our Help Center for more security tips[2].
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Q14 Look in the plaintext version of the email. In defanged format, what is the first URL extracted from the email?

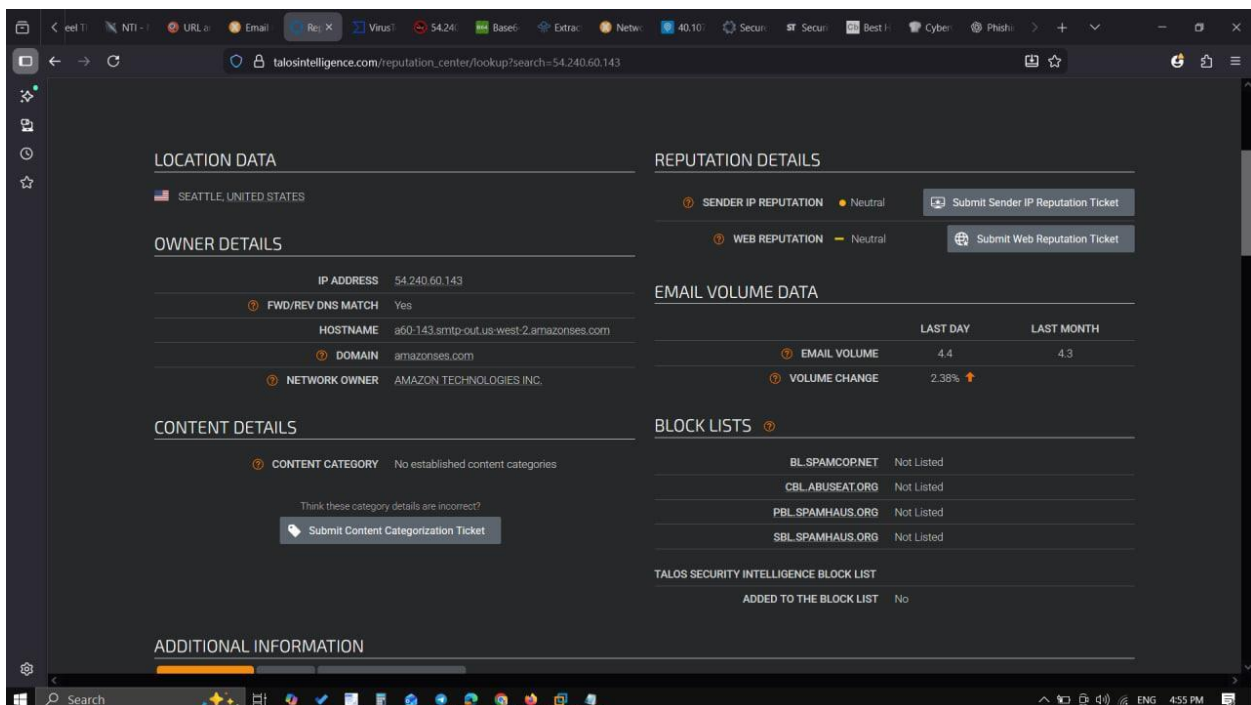
https://www.dropbox.com/l/ABCIzswTTJ9--CxR05fYXX35pPA-Y0m3PY/forgot_finish



Q15 Perform a Cisco Talos lookup on the base domain of the URL in the previous question.

What is its web reputation?

Neutral



Q16 [Yes or No] - After your analysis, is this email genuine?

Yes

Conclusion

The email is sent on behalf of **Dropbox** using **Amazon Simple Email Service (SES)**, which explains why Amazon-related domains appear in the headers.

- **SPF, DKIM, and DMARC authentication checks all pass**, confirming the email is authorized and not spoofed.
- The **From domain (dropbox.com)**, **Return-Path**, and **DKIM signing domain** are properly aligned, satisfying Dropbox's strict DMARC policy (p=REJECT).
- The **Message-ID** is generated by Amazon SES, while the **X-Dropbox-Message-ID** is an internal identifier used by Dropbox—this is expected behavior.
- All embedded links redirect to **official https://www.dropbox.com URLs**, with no suspicious redirects or mismatched domains.
- The content matches a **standard Dropbox password reset notification** and includes security guidance for users.

Final Verdict:

Legitimate Email – Authorized Dropbox Password Reset Notification

